

Early Warning Signals for bifurcations in multi-dimensional dynamical systems based on analysis of correlations in time series



Joshua Prettyman

NPL and
University of Reading

October 11, 2016

Background



MA - Pure Maths



Imperial College
London

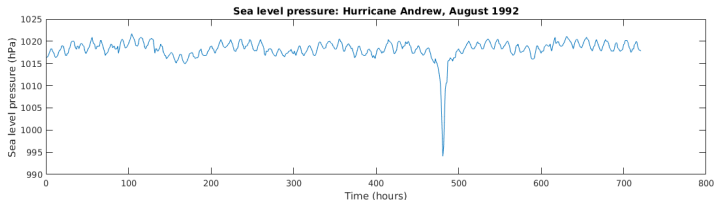
MRes - Solving PDEs



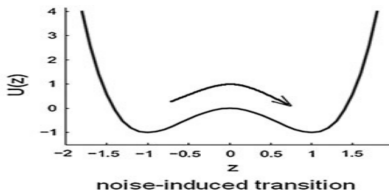
PhD - Time Series with Valerie

Purpose of the Project

- Time series may change in subtle ways before an obvious, extreme change, or **tipping point**.
- By detecting these changes we can predict the extreme change, in this we have an **Early Warning Signal**.
- By using a variety of possible predictors, we hope to create a **multi-purpose tool** for use with geophysical time series.

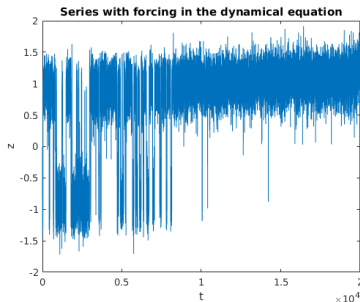


Tipping point example: Double well potential



- We may detect a transition from one state to another.
- The shape of the potential may change (bifurcation).
- The potential may shift.

$$\dot{z}(t) = -(z^4 - 2z^2)' + F(t) + \sigma\eta$$
$$F(t) = (5 \times 10^{-9})t^2$$



Tipping point detection: main ideas

Model fitting

- Use a window of data to parametrise a model.
- Look for trends in the changing shape of the model.

Finger-printing

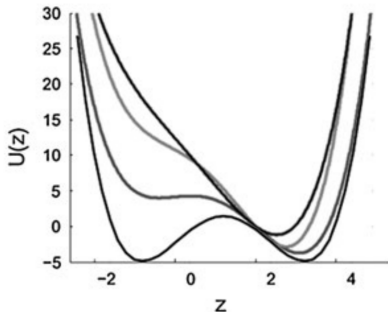
- Calculate some statistics in a sliding window.
- Look for trends.
- Expect that the dynamics will change when a threshold is reached.

Critical Slowing Down

When the system potential is undergoing a bifurcation:

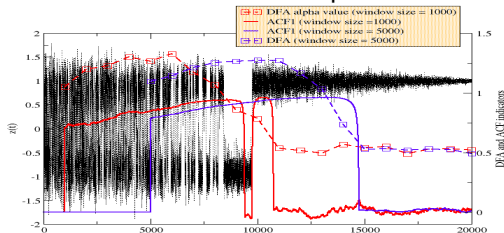
- The stable mode becomes less stable
- The system takes longer to return to the stable mode
- Eventually the system will take “forever” to return

Bifurcating potential function



Early Warning Signals

SDE model output



$$\dot{z}(t) = -(z^4 - 2z^2)' + \sigma(t)\eta$$
$$\sigma(t) = -0.00007t + 1.50045$$

Autocorrelation Function

- Choice of window size
- Choice of estimator

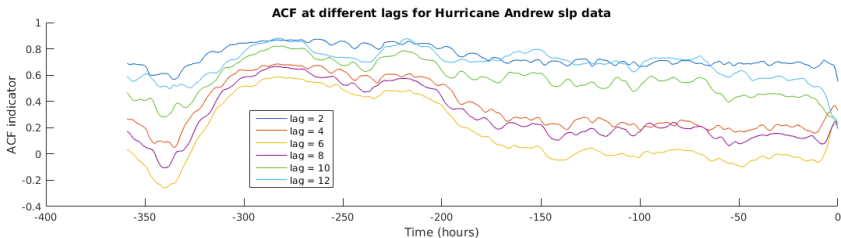
Detrended Fluctuation Analysis

- Choice of window size
- Choice of 'detrending'

Calculating ACF

- Calculate in a sliding window
- We use an estimator:

$$\text{ACF}_{(1)}([x_i]_1^N) = \frac{\frac{1}{N-1} \sum_{i=2}^N (x_{i-1} - \bar{x})(x_i - \bar{x})}{\text{Var}(x)}$$



Detrended Fluctuation Analysis

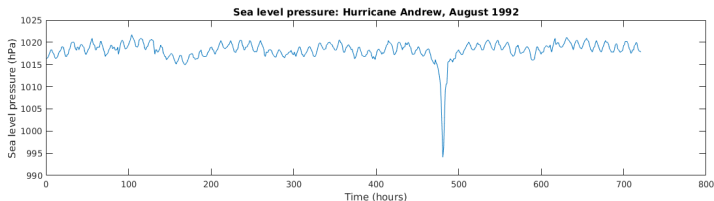
From [Kantelhardt et al., 2001]:

1. Y is the cumulative sum of the time series.
2. Divide Y into $\lfloor N/s \rfloor$ segments of length s .
3. Detrend each segment of Y by subtracting the order- n polynomial fit. This gives Y_s .
4. Calculate the variance of each Y_s , call this F_s . Then calculate

$$F^{(n)}(s) = \sqrt{\frac{1}{\lfloor N/s \rfloor} \sum F_s}$$

5. $F^{(n)}(s) \propto s^\alpha$. Thus, we do the above four steps for a range of s values, then fit an exponential to get a value for α .

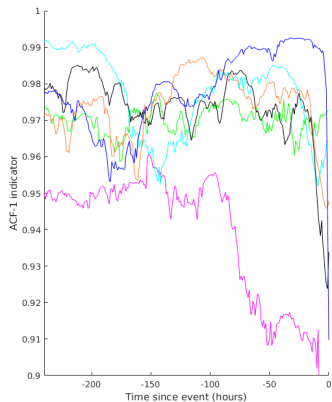
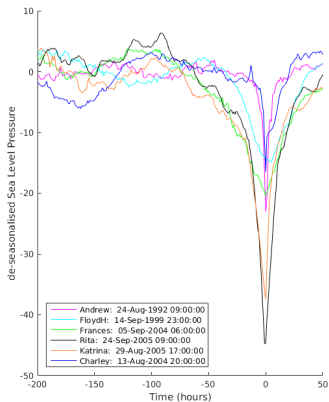
Tropical Cyclones



- An example of something dramatic happening suddenly
 - We expect a subtle change in memory as the cyclone approaches
-

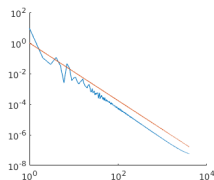
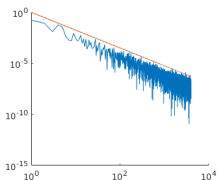
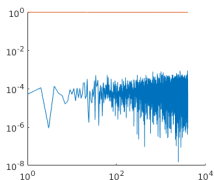
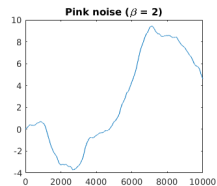
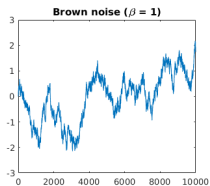
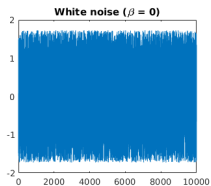
- We must decide what, where and when
- Cyclones -and the weather- are very complicated

First results - ACF

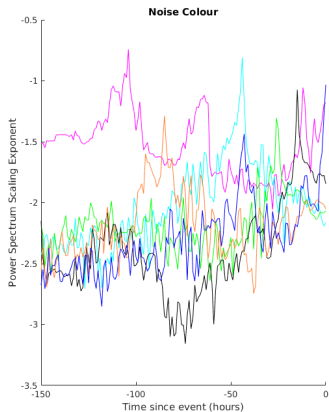
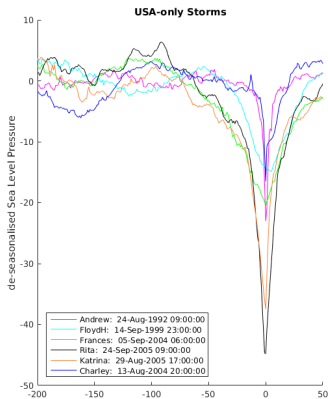


Noise colour

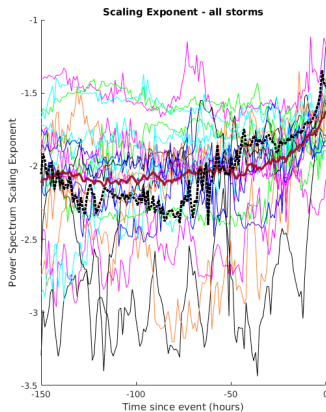
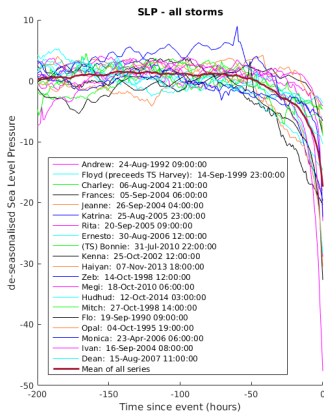
Different 'coloured' signals, and associated power spectra



First results - Noise colour



First results - Noise colour

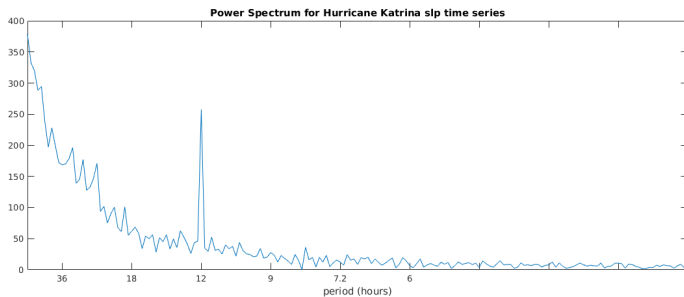


Removing oscillations

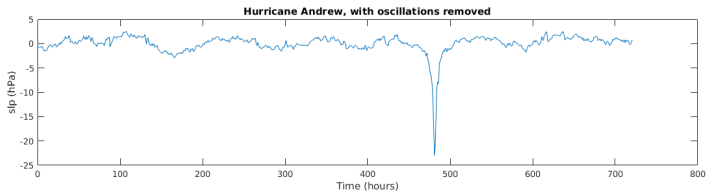
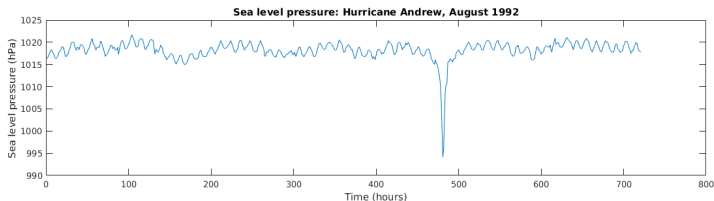
- “De-seasonalise”:

$$x(t) \rightarrow x_t - \text{mean}([x_i]) \quad i \equiv 0 \pmod{t}$$

- Subtract a composite of sine waves



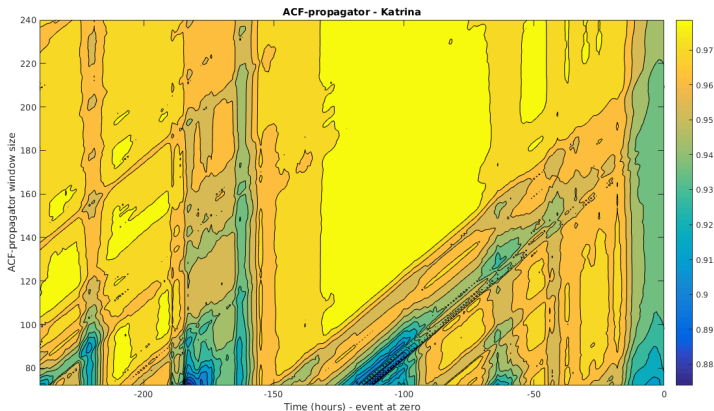
Removing oscillations



Sensitivity Analysis

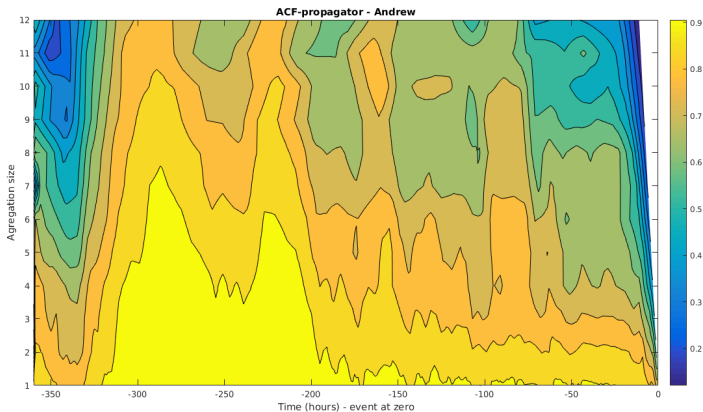
Changing the window size:

- At some point there must be a window size pay-off.



Sensitivity Analysis

Aggregation of data:



Multidimensional early warning signal

From [Williamson and Lenton, 2015]: In 1-D we estimate a using

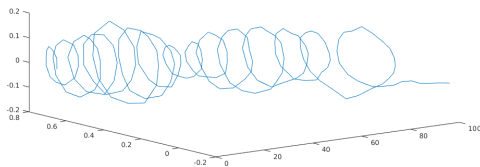
$$a = \frac{E(x_{t+1}x_t) - \mu^2}{E(x_t^2) - \mu^2} = b\sigma^{-1}$$

In the multivariate case we do the same:

$$A = \frac{E(\mathbf{x}_{t+1}\mathbf{x}_t) - \mu\mu^T}{E(\mathbf{x}_t^2) - \mu\mu^T} = B\Sigma^{-1}$$

Multidimensional early warning signal

SDE model output



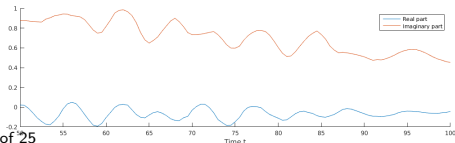
$$\dot{x} = y + \varepsilon^{(x)}$$

$$\dot{y} = \mu - x^2 + \varepsilon^{(y)}$$

$$\mu = 0.5 - 0.005t$$

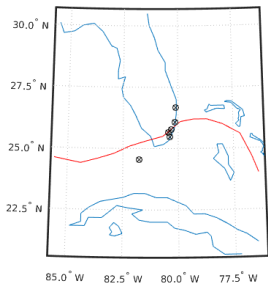
[Williamson and Lenton, 2015]

Eigenvalues of the Jacobian

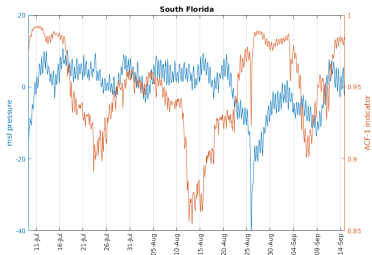


Multidimensional early warning signal

Path of Katrina



First PCA score, with ACF (240 point window)



Future research plans

- Revisit the ideas of multidimensional time series: geographically and in state-space.
- Use ridge-regression methods for this purpose.
- Create a stochastic (SDE/ARMA) model to fit the data.
- Aim for a precise future prediction.
- Construct a more suitable estimator for ACF.
- Remove oscillations from pressure data.

Learning during the first year

- Interpreting Power spectra of time series
- MatLab, NetCDF data processing
- Sensitivity analysis
- Questioning methods
- Thinking about null hypotheses
- Questions of probability and uncertainty
- Giving presentations

Workshops and Conferences

Attended:

- Industry-focussed mathematical modelling Graduate Camp (March 2016, Oxford)
- Mathematics of Planet Earth Jamboree (March 2016, Reading)

Planned:

- Time-series analysis in Environmental Science and Application to Climate Change (November 2016, Tromsø)

Select Bibliography

- Held and Kleinen, 2004. Detection of climate system bifurcations by degenerate fingerprinting. *Geophysical Research Letters*, 31(23).
- Kantelhardt et al., 2001. Detecting long-range correlations with de-trended fluctuation analysis. *Physica A: Statistical Mechanics*, 295(3):441454.
- Livina et al., 2011. Changing climate states and stability: from pliocene to present. *Climate dynamics*, 37(11-12):24372453.
- Livina and Lenton, 2007. A modified method for detecting incipient bifurcations in a dynamical system. *Geophysical Research Letters*, 34(3).
- Williamson and Lenton, 2015. Detection of bifurcations in noisy coupled systems from multiple time series. *Chaos*, 25(3):036407.

Thank you!